

UC DAVIS HEALTH

Drug Dialyzability Reference

Complete Guide: Conventional HD, High-Flux HD, CRRT

August 2026 | 200+ Drugs | New & International Medications

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FACTORS AFFECTING DRUG DIALYZABILITY

Factor	High Dialyzability	Low Dialyzability
Molecular Weight	< 500 Da	> 1000 Da
Protein Binding	< 20%	> 80%
Volume of Distribution	< 1 L/kg	> 2 L/kg
Water Solubility	High	Low (lipophilic)

CRRT MODALITIES

Modality	Full Name	Mechanism	Drug Clearance
CVVH	Continuous Veno-Venous Hemofiltration	Convection	HIGH - Middle molecules
CVVHD	Continuous Veno-Venous Hemodialysis	Diffusion	MODERATE - Small molecules
CVVHDF	Continuous Veno-Venous Hemodiafiltration	Both	HIGHEST - Broad range
SCUF	Slow Continuous Ultrafiltration	Ultrafiltration only	MINIMAL - Fluid only

QUICK REFERENCE

ALWAYS GIVE POST-HEMODIALYSIS:

Aminoglycosides, Vancomycin, Beta-lactams, Gabapentin/Pregabalin, Acyclovir, Atenolol, Sotalol, Sitagliptin, Fluconazole

NEVER SUPPLEMENT:

Warfarin, Phenytoin, Valproic Acid, Digoxin, Amiodarone, SSRIs, TCAs, Benzodiazepines, Statins, Insulin

CONTRAINDICATED IN ESRD:

Metformin (lactic acidosis), Lithium (toxicity), Dabigatran (accumulation), Morphine (metabolites), NSAIDs, Sulfonylureas

ANTIBIOTICS & ANTIMICROBIALS

Drug	Class	MW	PB	Vd	HD	HF	CRRT	Dosing
Amikacin	Aminoglycoside	586	0%	0.25	Y	Y	Y	Give AFTER HD; monitor levels
Gentamicin	Aminoglycoside	478	0%	0.25	Y	Y	Y	Give AFTER HD; monitor levels
Tobramycin	Aminoglycoside	468	0%	0.3	Y	Y	Y	Give AFTER HD; monitor levels
Vancomycin	Glycopeptide	1449	50%	0.7	M	Y	Y	15-20 mg/kg post-HD
Teicoplanin	Glycopeptide	1879	90%	0.1	N	M	M	6mg/kg post-HD q48h
Daptomycin	Lipopeptide	1620	90%	0.1	N	M	M	6-8 mg/kg post-HD q48h
Linezolid	Oxazolidinone	337	31%	0.6	M	Y	Y	No supplement; monitor CBC
Tedizolid	Oxazolidinone	352	70-90%	1.5	N	N	M	200mg daily
Cefazolin	Cephalosporin 1st	455	80%	0.2	M	Y	Y	1g post-HD q24h
Cefepime	Cephalosporin 4th	571	20%	0.3	Y	Y	Y	1-2g post-HD
Ceftazidime	Cephalosporin 3rd	547	10%	0.2	Y	Y	Y	1g post-HD
Ceftriaxone	Cephalosporin 3rd	555	90%	0.15	N	N	N	No supplement; hepatic
Ceftolozane	Cephalosporin 5th	606	20%	0.3	Y	Y	Y	1.5g post-HD
Meropenem	Carbapenem	438	2%	0.3	Y	Y	Y	1g post-HD
Imipenem-Cilastatin	Carbapenem	523	20%	0.3	Y	Y	Y	500mg post-HD
Doripenem	Carbapenem	420	10%	0.2	Y	Y	Y	500mg post-HD
Ertapenem	Carbapenem	523	85%	0.1	N	M	M	500mg post-HD q48h
Piperacillin-Tazobactam	Penicillin+BLI	517	30%	0.3	Y	Y	Y	4.5g post-HD
Ampicillin-Sulbactam	Penicillin+BLI	491	25%	0.2	Y	Y	Y	3g post-HD
Amoxicillin-Clavulanate	Penicillin+BLI	428	25%	0.3	Y	Y	Y	1g post-HD
Nafcillin	Penicillin	415	90%	0.3	N	N	N	No supplement; hepatic
Ciprofloxacin	Fluoroquinolone	332	30%	2.5	M	M	M	500mg post-HD
Levofloxacin	Fluoroquinolone	362	30%	1.0	M	Y	Y	500mg post-HD
Moxifloxacin	Fluoroquinolone	402	45%	2.0	N	N	N	No supplement; hepatic
Delafoxacin	Fluoroquinolone	478	90%	0.4	N	N	M	300mg q48h
Azithromycin	Macrolide	749	50%	31	N	N	N	No supplement
Clarithromycin	Macrolide	748	70%	4.0	N	N	N	No supplement; hepatic
Doxycycline	Tetracycline	445	90%	1.0	N	N	N	No supplement; hepatic
Minocycline	Tetracycline	458	75%	1.0	N	N	N	No supplement; hepatic
Tigecycline	Glycylcycline	586	70%	8.0	N	N	N	No supplement; large Vd
Colistin	Polymyxin	1745	50%	0.1	M	Y	Y	2-3 MU post-HD
Fosfomycin	Phosphonic Acid	260	0%	0.4	Y	Y	Y	4g post-HD
Rifampin	Rifamycin	823	85%	1.0	N	N	N	No supplement; hepatic
Isoniazid	Antimycobacterial	137	0%	0.7	Y	Y	Y	300mg post-HD daily
Pyrazinamide	Antimycobacterial	123	0%	0.7	Y	Y	Y	25-30 mg/kg post-HD
Ethambutol	Antimycobacterial	204	10%	1.5	M	M	M	15-25 mg/kg post-HD
Bedaquiline	Diarylquinoline	556	99%	166	N	N	N	400mg weekly

Fluconazole	Azole	307	12%	0.7	Y	Y	Y	400mg post-HD daily
Voriconazole	Azole	350	58%	4.0	N	N	N	No supplement; hepatic
Posaconazole	Azole	701	98%	3.0	N	N	N	No supplement; hepatic
Isavuconazole	Azole	724	97%	0.4	N	N	N	372mg q24h
Echinocandins	Echinocandin	>1200	>95%	0.1	N	N	N	No supplement
Amphotericin B	Polyene	924	90%	4.0	N	N	N	No supplement
Acyclovir	Antiviral	225	15%	0.7	Y	Y	Y	5 mg/kg post-HD daily
Valacyclovir	Antiviral	325	15%	0.7	Y	Y	Y	500mg post-HD daily
Ganciclovir	Antiviral	256	2%	0.7	Y	Y	Y	2.5 mg/kg post-HD 3x/wk
Valganciclovir	Antiviral	355	2%	0.7	Y	Y	Y	450mg post-HD 3x/wk
Oseltamivir	Antiviral	313	42%	0.3	M	M	M	30mg post-HD daily
Remdesivir	Antiviral	603	90%	0.4	N	N	N	No supplement; hepatic

CARDIOVASCULAR MEDICATIONS

Drug	Class	MW	PB	Vd	HD	HF	CRRT	Dosing
Lisinopril	ACE Inhibitor	406	0%	1.3	Y	Y	Y	5-10mg post-HD daily
Enalapril	ACE Inhibitor	377	50%	1.0	M	Y	Y	5mg post-HD daily
Ramipril	ACE Inhibitor	417	75%	1.0	N	M	M	2.5mg daily
Losartan	ARB	423	98%	0.1	N	N	N	No supplement; hepatic
Valsartan	ARB	436	95%	0.2	N	N	N	No supplement; hepatic
Amlodipine	CCB	409	93%	21	N	N	N	No supplement; hepatic
Nifedipine	CCB	347	90%	1.0	N	N	N	No supplement; hepatic
Diltiazem	CCB	451	80%	5.0	N	N	N	No supplement; hepatic
Verapamil	CCB	455	90%	5.0	N	N	N	No supplement; hepatic
Metoprolol	Beta-Blocker	266	12%	5.0	M	M	M	Titrate to HR
Atenolol	Beta-Blocker	267	5%	1.0	Y	Y	Y	25-50mg post-HD daily
Carvedilol	Beta-Blocker	407	98%	2.0	N	N	N	No supplement; hepatic
Bisoprolol	Beta-Blocker	315	88%	3.0	N	N	N	5-10mg daily
Furosemide	Loop Diuretic	331	99%	0.2	N	N	N	Give IV during HD
Torsemide	Loop Diuretic	349	99%	0.1	N	N	N	No supplement; hepatic
Spironolactone	K-Sparing	417	90%	1.5	N	N	N	AVOID (hyperkalemia)
Eplerenone	K-Sparing	416	50%	1.0	N	M	M	AVOID (hyperkalemia)
Digoxin	Cardiac Glycoside	781	25%	7.0	N	N	N	Monitor levels
Amiodarone	Antiarrhythmic	646	97%	66	N	N	N	No supplement; hepatic
Sotalol	Antiarrhythmic	273	0%	2.0	Y	Y	Y	40-80mg post-HD
Dofetilide	Antiarrhythmic	442	60%	3.0	M	M	M	Contraindicated CrCl<20
Warfarin	Anticoagulant	309	99%	0.1	N	N	N	Monitor INR
Apixaban	DOAC	460	87%	0.3	N	N	N	AVOID if ESRD
Rivaroxaban	DOAC	436	95%	0.5	N	N	N	AVOID if CrCl<15
Dabigatran	DOAC	472	35%	0.7	M	Y	Y	CONTRAINDICATED in ESRD
Edoxaban	DOAC	549	55%	1.0	M	M	M	AVOID if CrCl<15
Heparin (UFH)	Anticoagulant	15000	80%	0.05	N	N	N	Stop 4h before HD
Enoxaparin	LMWH	4500	0%	0.1	N	N	N	20-30mg daily

NEUROLOGIC & PSYCHIATRIC MEDICATIONS

Drug	Class	MW	PB	Vd	HD	HF	CRRT	Dosing
Gabapentin	Anticonvulsant	172	0%	1.0	Y	Y	Y	100-300mg post-HD daily
Pregabalin	Anticonvulsant	160	0%	0.5	Y	Y	Y	25-75mg post-HD daily
Levetiracetam	Anticonvulsant	172	0%	0.7	Y	Y	Y	250-500mg post-HD daily
Valproic Acid	Anticonvulsant	145	90%	0.2	N	N	N	Monitor levels
Carbamazepine	Anticonvulsant	237	75%	1.5	N	N	N	No supplement; hepatic
Lamotrigine	Anticonvulsant	257	55%	1.0	N	N	N	No supplement; hepatic
Topiramate	Anticonvulsant	340	15%	0.8	M	Y	Y	25-50mg post-HD daily
Phenytoin	Anticonvulsant	253	90%	0.7	N	N	N	Monitor free levels
Sertraline	SSRI	325	98%	20	N	N	N	No supplement; hepatic
Fluoxetine	SSRI	310	95%	45	N	N	N	No supplement; hepatic
Paroxetine	SSRI	330	95%	8.0	N	N	N	No supplement; hepatic
Citalopram	SSRI	325	80%	15	N	N	N	No supplement; hepatic
Escitalopram	SSRI	325	80%	15	N	N	N	No supplement; hepatic
Venlafaxine	SNRI	278	30%	7.0	M	M	M	37.5-75mg daily
Duloxetine	SNRI	298	96%	16	N	N	N	No supplement; hepatic
Bupropion	NDRI	240	85%	20	N	N	N	No supplement; hepatic
Mirtazapine	Tetracyclic	266	85%	4.0	N	N	N	No supplement; hepatic
Trazodone	SARI	372	90%	1.0	N	N	N	No supplement; hepatic
Quetiapine	Antipsychotic	384	83%	10	N	N	N	No supplement; hepatic
Olanzapine	Antipsychotic	313	93%	16	N	N	N	No supplement; hepatic
Risperidone	Antipsychotic	411	90%	1.5	N	N	N	No supplement; hepatic
Aripiprazole	Antipsychotic	448	99%	5.0	N	N	N	No supplement; hepatic
Lurasidone	Antipsychotic	499	99%	6.0	N	N	N	No supplement; hepatic
Brexiprazole	Antipsychotic	437	99%	3.0	N	N	N	No supplement; hepatic
Cariprazine	Antipsychotic	474	99%	20	N	N	N	No supplement; hepatic
Lithium	Mood Stabilizer	7	0%	1.0	Y	Y	Y	AVOID in ESRD (toxic)
Lorazepam	Benzodiazepine	322	85%	1.0	N	N	N	No supplement; hepatic
Alprazolam	Benzodiazepine	309	80%	1.0	N	N	N	No supplement; hepatic
Clonazepam	Benzodiazepine	316	85%	3.0	N	N	N	No supplement; hepatic
Zolpidem	Hypnotic	308	93%	1.0	N	N	N	No supplement; hepatic

ENDOCRINE & DIABETES MEDICATIONS

Drug	Class	MW	PB	Vd	HD	HF	CRRT	Dosing
Metformin	Biguanide	130	0%	1.0	Y	Y	Y	CONTRAINDICATED (lactic acidosis)
Glipizide	Sulfonylurea	446	99%	0.1	N	N	N	No supplement; hepatic
Glyburide	Sulfonylurea	495	99%	0.2	N	N	N	AVOID (hypoglycemia)
Glimepiride	Sulfonylurea	491	99%	0.2	N	N	N	AVOID (hypoglycemia)
Sitagliptin	DPP-4 Inhibitor	408	38%	1.0	Y	Y	Y	25mg daily (50mg post-HD 3x/wk)
Linagliptin	DPP-4 Inhibitor	473	99%	2.0	N	N	N	5mg daily; no adjustment
Saxagliptin	DPP-4 Inhibitor	316	0%	2.0	M	M	M	2.5mg daily
Alogliptin	DPP-4 Inhibitor	340	0%	4.0	M	M	M	6.25mg daily
Empagliflozin	SGLT2 Inhibitor	451	68%	2.0	N	N	N	NOT rec if eGFR<20
Dapagliflozin	SGLT2 Inhibitor	409	91%	2.0	N	N	N	NOT rec if eGFR<25
Canagliflozin	SGLT2 Inhibitor	445	99%	1.0	N	N	N	NOT rec if eGFR<30
Insulin (all)	Hormone	5808	0%	0.2	N	N	N	Monitor glucose
Levothyroxine	Thyroid	777	99%	0.2	N	N	N	Monitor TSH
Prednisone	Corticosteroid	360	70%	1.0	N	N	N	No supplement; hepatic
Methylprednisolone	Corticosteroid	375	78%	1.0	N	N	N	No supplement; hepatic

ANALGESICS

Drug	Class	MW	PB	Vd	HD	HF	CRRT	Dosing
Acetaminophen	Analgesic	152	20%	1.0	Y	Y	Y	650mg q6h PRN
Ibuprofen	NSAID	206	99%	0.1	N	N	N	AVOID (bleeding, renal)
Naproxen	NSAID	231	99%	0.2	N	N	N	AVOID (bleeding, renal)
Ketorolac	NSAID	256	99%	0.2	N	N	N	CONTRAINDICATED in ESRD
Celecoxib	COX-2 Inhibitor	382	97%	1.0	N	N	N	AVOID in ESRD
Tramadol	Opioid	264	20%	3.0	M	M	M	50mg post-HD q12h PRN
Morphine	Opioid	286	35%	3.0	M	M	M	AVOID (metabolite accumulation)
Hydromorphone	Opioid	286	30%	2.0	M	M	M	1-2mg post-HD q12h PRN
Oxycodone	Opioid	316	45%	2.0	M	M	M	2.5-5mg post-HD q12h PRN
Fentanyl	Opioid	337	80%	4.0	N	N	N	No supplement; hepatic
Methadone	Opioid	311	90%	5.0	N	N	N	No supplement; hepatic
Buprenorphine	Opioid	468	96%	3.0	N	N	N	No supplement; hepatic
Codeine	Opioid	300	7%	3.0	M	M	M	AVOID (variable metabolism)
Hydrocodone	Opioid	300	20%	2.0	M	M	M	2.5-5mg post-HD q12h PRN

NEW DRUGS (2020-2026)

Drug	Class	Year	MW	Dialyzable	Key Point
Semaglutide	GLP-1 Agonist	2021	4114	N	Large molecule; hepatic
Tirzepatide	GIP/GLP-1 Agonist	2022	4806	N	Large molecule; not for ESRD
Inclisiran	PCSK9 siRNA	2021	16000	N	Large molecule; q6months
Evolocumab	PCSK9 Inhibitor	2020	146000	N	Large molecule; q2-4weeks
Alirocumab	PCSK9 Inhibitor	2020	146000	N	Large molecule; q2-4weeks
Bempedoic Acid	ACL Inhibitor	2020	447	N	Hepatic; no adjustment
Lefamulin	Pleuromutilin	2020	557	N	Hepatic; 600mg q12h
Omadacycline	Tetracycline	2020	477	N	Hepatic; no adjustment
Eravacycline	Fluorocycline	2020	544	N	Hepatic; no adjustment
Cefiderocol	Siderophore Cephalosporin	2020	823	M	2g q12h; adjust for CrCl
Plazomicin	Aminoglycoside	2020	592	Y	Give post-HD; monitor levels
Delaflaxacin	Fluoroquinolone	2020	478	N	Hepatic; 300mg q48h
Lasmiditan	5-HT _{1F} Agonist	2020	344	N	Hepatic; no adjustment
Ubrogepant	CGRP Antagonist	2020	454	N	Hepatic; no adjustment
Rimegepant	CGRP Antagonist	2021	517	N	Hepatic; 75mg q48h
Atogepant	CGRP Antagonist	2022	517	N	Hepatic; 10-60mg daily
Deutetrabenazine	VMAT2 Inhibitor	2020	376	N	Hepatic; no adjustment
Pimavanserin	5-HT _{2A} Antagonist	2020	421	N	Hepatic; no adjustment

UC Davis Health - Renal Services
Drug Dialyzability Reference - Complete Edition
Updated: August 2026

For clinical use only. Verify all calculations and dosing recommendations.
References: KDIGO 2025, KDOQI 2025, FDA/EMA Prescribing Information, UpToDate, Lexicomp

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